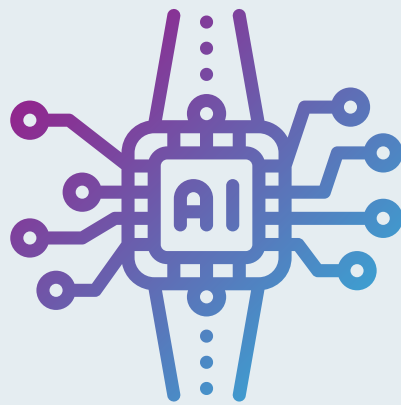


DEEP REINFORCEMENT LEARNING FOR REAL-TIME STRATEGY GAMES: MICRORTS CASE STUDY



Author: **Mathis DELSART** Supervisor: **Eric PIETTE** Academic Year: **2025-2026**
Achille MORENVILLE



RTS Games: What Are They?

- Decision-making in real time (no turns)
- Resource management & unit production
- Micro- and macro-level tactical control
- Massive action space (every unit can act each tick)

CONTEXT

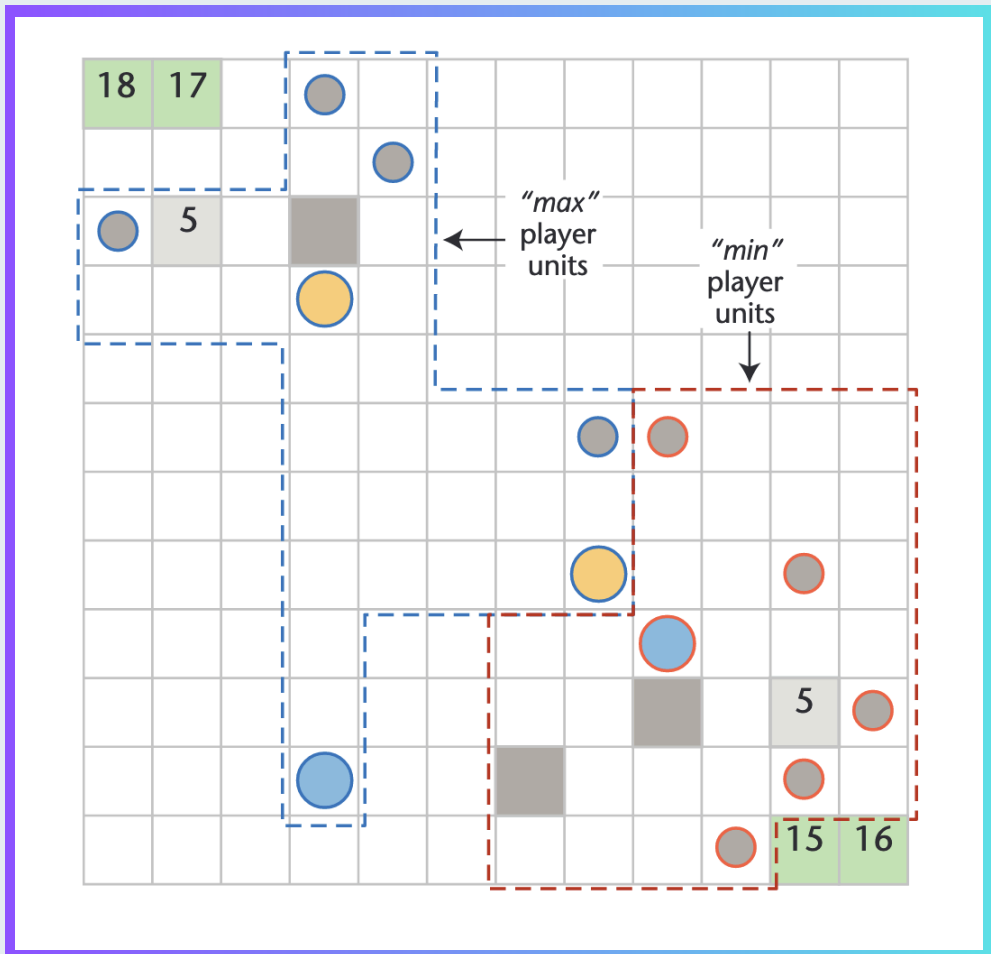
- MicroRTS**: minimalist RTS environment for AI research
- NxM grid, units & buildings - destroy all enemy forces
- Benchmark** for new AI methods & annual competition

MOTIVATION

- Challenges**: large action space, sparse rewards, real-time decisions,...
- Contribution**: Advancing Deep RL for RTS research
- Competitive & gameplay relevance**

CURRENTLY

- Deep RL agents now **surpass** scripted & hybrid (script+ML) bots
- Reference: **RAISocketAI**, the current top Deep RL bot



MicroRTS Illustration: Ontañón et al., The First MicroRTS Artificial Intelligence Competition, AI Magazine, 2018



MicroRTS Logo: <https://github.com/Farama-Foundation/MicroRTS/blob/master/microrts-text.png>

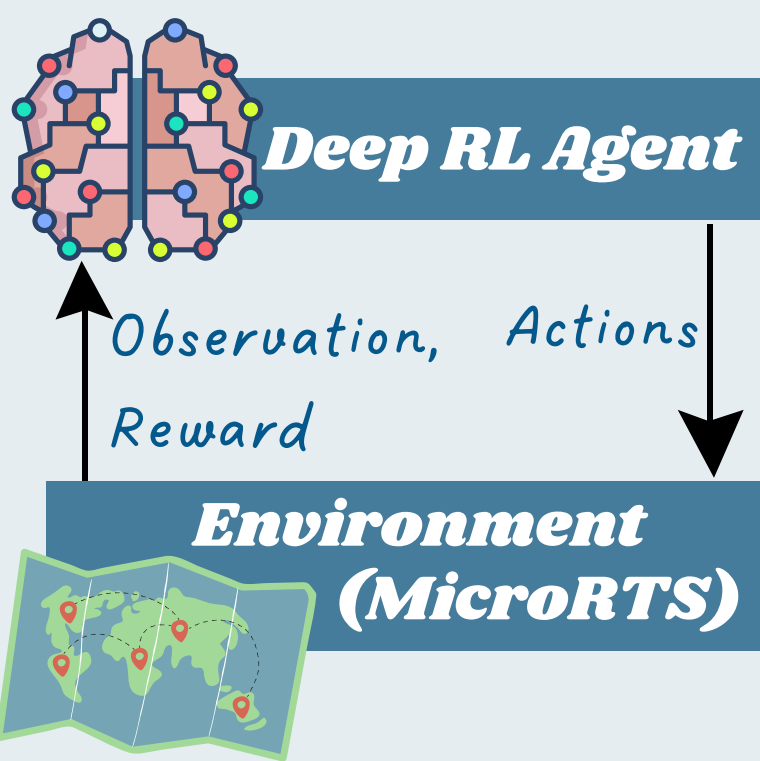
GOAL

- Develop a **state-of-the-art** Deep RL agent for MicroRTS
- Submit to **MicroRTS Competition (IEEE 2026 CoG)**

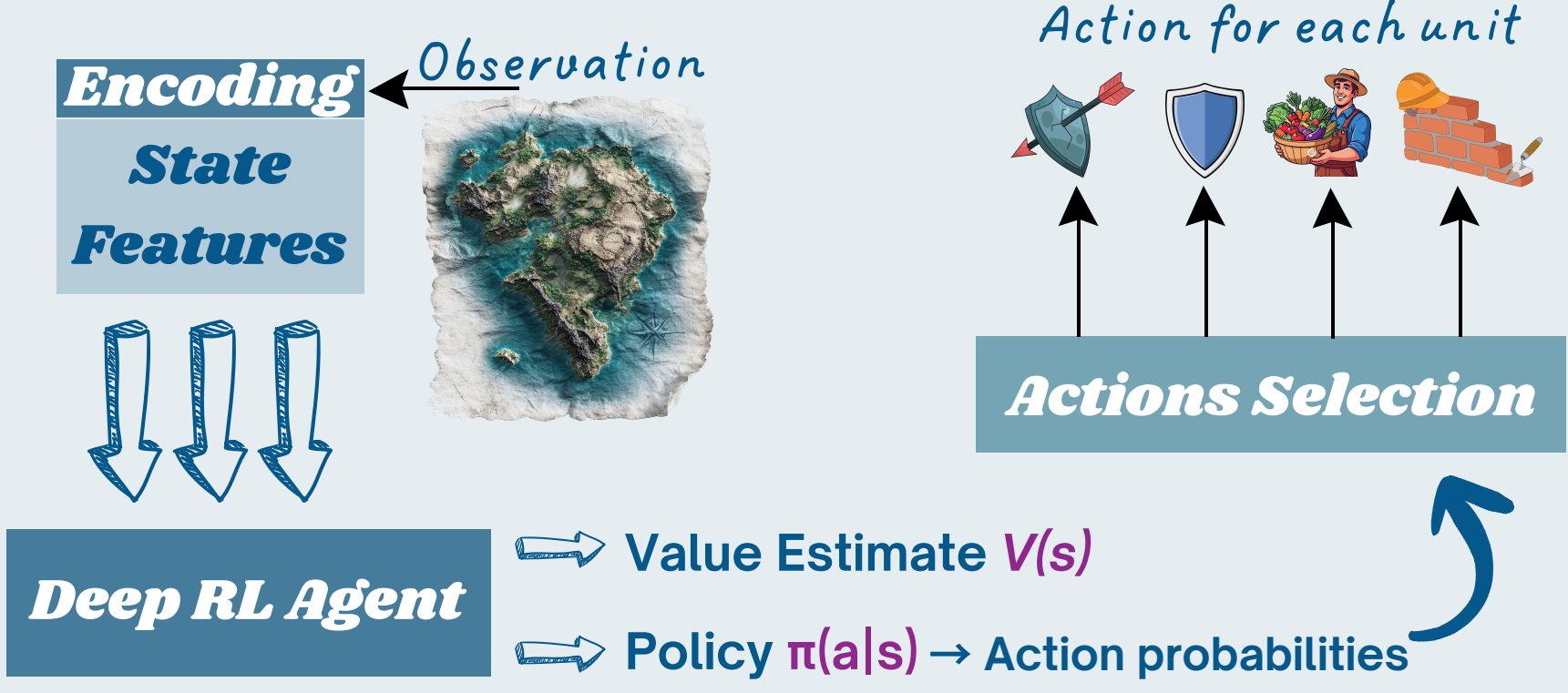
WHY DEEP RL ?

- Learns **complex strategies** through **self-play**
- Learns **long-term strategy** from **sparse rewards**
- Can **generalize** to **unseen opponents**

RL Interaction Loop



Deep RL Model Inference



Approach

LITERATURE



IMPLEMENTATION



Past competition winners provide **open-source code**, enabling to:

- Relaunch competitions** with previous winners as **benchmarks**
- Evaluate** our agent **against proven strategies**
- Test** across **different game scenarios and maps**
- Analyze** *strengths & weaknesses* systematically

EVALUATION

